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Related Literature Review

Proposed Project: Papaya Maturity Classification Using Machine Learning

1. Introduction and Objective

The objective of this review is to build a strong foundation for the proposed project, Papaya Maturity Classification Using Machine Learning, by systematically examining existing peer-reviewed research, identifying current trends in fruit-maturity classification, and locating the specific gaps that the proposed project is positioned to address. Ten related academic sources were identified through targeted searches of peer-reviewed journals and IEEE conference proceedings, selected for their direct relevance to machine learning and deep learning approaches to fruit maturity and ripeness classification. For each source, this document presents a full citation, a concise summary covering the study's objective, methodology, and key findings, and an explanation of how the source relates to and informs the proposed project. Read together, these sources establish the current state of the art, surface the methodological patterns and limitations recurring across the field, and clarify where the proposed project can make a meaningful and well-justified contribution.

2. Reviewed Sources

2.1 Maturity Status Classification of Papaya Fruits Based on Machine Learning and Transfer Learning Approach

Behera, S. K., Rath, A. K., & Sethy, P. K. (2020). Maturity status classification of papaya fruits based on machine learning and transfer learning approach. Information Processing in Agriculture, 8(2), 244–250 (published online 20 May 2020; issue publication 2021). https://doi.org/10.1016/j.inpa.2020.05.003

Summary: This study set out to solve the problem of slow, subjective, and destructive manual grading of papaya maturity in the fruit-packaging industry. The researchers photographed 300 papaya fruits evenly split across three maturity classes (immature, partially mature, and mature) and compared two non-destructive classification pipelines. The first used handcrafted image features — local binary pattern (LBP), histogram of oriented gradients (HOG), and gray-level co-occurrence matrix (GLCM) — paired with classical classifiers (k-nearest neighbor, support vector machine, and Naïve Bayes). The second applied transfer learning using seven pretrained convolutional networks (AlexNet, GoogLeNet, VGG16, VGG19, and three ResNet variants). The weighted KNN classifier using HOG features and the fine-tuned VGG19 model both achieved 100% classification accuracy, with VGG19 outperforming a prior benchmark method by 6 percentage points while training in under two minutes.

Project Relevance: This paper is the closest direct precedent to the proposed project: it tackles the identical problem (three-class papaya maturity classification from external images) using both classical machine learning and transfer learning, providing a ready-made benchmark of feature sets, classifiers, and CNN backbones to replicate or improve upon. Its principal limitation — a small, single-location, 300-image dataset — is a gap the proposed project can directly address by collecting a larger and more varied image set.

2.2 Multimodal Deep Learning and Visible-Light and Hyperspectral Imaging for Fruit Maturity Estimation

Garillos-Manliguez, C. A., & Chiang, J. Y. (2021). Multimodal deep learning and visible-light and hyperspectral imaging for fruit maturity estimation. Sensors, 21(4), 1288. https://doi.org/10.3390/s21041288

Summary: The researchers observed that most fruit-maturity classifiers collapse ripeness into only two coarse classes (ripe/unripe) and rely on a single (unimodal) imaging source, potentially discarding internal-quality information. To address this, they built a laboratory database of papaya fruit captured with both ordinary RGB cameras and hyperspectral sensors (400–900 nm) across six finer-grained maturity stages. Seven CNN architectures (including AlexNet, VGG16/19, ResNet50, and MobileNet variants) were adapted into multimodal versions that fused RGB and hyperspectral features through late fusion. The multimodal models consistently outperformed their single-modality counterparts, reaching F1-scores up to 0.90 and a top-2 error rate of only 1.45% across the six maturity stages.

Project Relevance: This study demonstrates the accuracy ceiling achievable when maturity classification is enriched with spectral (not just visible-light) data and finer-grained class labels, giving the proposed project a useful upper-bound reference point. It also justifies why an RGB-only, three- or four-class design — more typical of what a smallholder-oriented or thesis-scale project can realistically deploy — represents a practical trade-off between accuracy and cost, since hyperspectral hardware and six-way labelling are not feasible for most field or market settings.

2.3 Detection of Papaya Ripeness Using Deep Learning Approach

Gayathri, S., Ujwala, T. U., Vinusha, C. V., Pauline, N. R., & Tharunika, D. B. (2021). Detection of papaya ripeness using deep learning approach. In 2021 Third International Conference on Inventive Research in Computing Applications (ICIRCA) (pp. 1755–1758). IEEE. https://doi.org/10.1109/ICIRCA51532.2021.9544902

Summary: Motivated by rising papaya production volumes and the misclassification risk of manual grading, this conference paper proposes a straightforward convolutional neural network (CNN) pipeline for classifying papaya ripeness into ripe, unripe, and partially ripe categories using

a 300-image dataset. Unlike earlier machine-learning approaches that depend on manually engineered colour or texture descriptors, the CNN learns discriminative features directly from raw images. The authors report that the deep-learning pipeline classifies ripeness effectively without any manual feature-extraction step, positioning end-to-end CNNs as a simpler and more automatic alternative to classical machine-learning pipelines for this task.

Project Relevance: As a minimal, proof-of-concept CNN baseline for exactly the three-class papaya ripeness problem, this paper offers a useful architectural starting point and a reminder of the field's recurring small-dataset limitation. It reinforces the case that the proposed project's contribution can lie in scaling up the dataset, testing robustness across varied lighting and backgrounds, and rigorously benchmarking CNN performance rather than reinventing the basic classification concept.

2.4 Prediction of the Ripening Stages of Papayas Using Discriminant Analysis and Support Vector Machine Algorithms

Zulkifli, N., Hashim, N., Harith, H. H., Mohamad Shukery, M. F., & Onwude, D. I. (2022). Prediction of the ripening stages of papayas using discriminant analysis and support vector machine algorithms. Journal of the Science of Food and Agriculture, 102(8), 3266–3276. https://doi.org/10.1002/jsfa.11669

Summary: This study addressed the gap between what a fruit looks like externally and what is actually happening internally as it ripens, since visual grading alone does not capture biochemical quality. Colour features were extracted from papaya images across four colour spaces (RGB, normalized RGB, HSV, and L*a*b*) at five distinct ripening stages, and correlated with mechanical firmness, moisture content, total soluble solids, and pH measured destructively on the same fruit. Linear discriminant analysis and support vector machine classifiers were then trained to predict ripening stage from the colour features alone. Mean colour values proved to be strong predictors of internal quality ($R^2 \geq 0.90$ for total soluble solids), and the classifiers reliably distinguished the five ripening stages from external appearance.

Project Relevance: By statistically linking external colour features to internal biochemical ripeness indicators, this paper provides scientific grounding for the core assumption behind any image-based papaya maturity classifier: that what a camera sees is a valid proxy for true internal ripeness. This is directly useful for justifying the proposed project's labelling protocol (e.g., tying image-based maturity classes to a verifiable ground truth) and for considering finer, five-stage labelling as an alternative to a simple three-class scheme.

2.5 Papaya Fruit Maturity Estimation Using Wavelet and ConvNET

Ratha, A. K., Barpanda, N. K., Sethy, P. K., & Behera, S. K. (2023). Papaya fruit maturity estimation using wavelet and ConvNET. Ingénierie des Systèmes d'Information, 28(1), 175–181. https://doi.org/10.18280/isi.280119

Summary: The authors noted that prior papaya-classification studies typically rely on either deep CNN features or handcrafted spectral/texture features, but rarely combine both, potentially leaving complementary information unused. Using the same 300-image, three-class dataset structure common in this research group's earlier work, they extracted deep features from a fine-tuned VGG16 network and, separately, 20 coefficients from a two-level Discrete Wavelet Transform (DWT) of each image. The two feature sets were concatenated (1,114 total features) and classified with a linear support vector machine. VGG16 features alone reached 96.7% accuracy, DWT features alone reached 75.7%, but the fused VGG16+DWT model achieved the best result of the three: 98% accuracy with a perfect AUC of 1.0.

Project Relevance: This paper offers a concrete, tested technique — hybrid deep-feature and wavelet-feature fusion — that the proposed project could adopt or use as a comparison model to push classification accuracy beyond a single CNN backbone. Its repeated use of the same small, single-source dataset across multiple papers from the same research group further underscores dataset diversity as a persistent, addressable gap in papaya-specific maturity-classification research.

2.6 Fruit Ripeness Classification: A Survey

Rizzo, M., Marcuzzo, M., Zangari, A., Gasparetto, A., & Albarelli, A. (2023). Fruit ripeness classification: A survey. Artificial Intelligence in Agriculture, 7, 44–57. https://doi.org/10.1016/j.aiia.2023.02.004

Summary: Recognizing that fruit-ripeness research is scattered across many fruit species, sensing modalities, and modelling paradigms, this survey systematically reviews peer-reviewed literature published between 2014 and 2022. It organizes existing work into statistical, classical machine-learning, and deep-learning approaches, and catalogues the public datasets and sensing technologies used (visible-light imaging, hyperspectral and multispectral imaging, fluorescence imaging, and electronic noses). The authors find that deep-learning methods, particularly CNNs, increasingly dominate top-performing systems because they remove the need for hand-engineered features, and that non-destructive spectral sensing is a fast-growing complement to plain visible imaging. They also note that standardized, publicly available benchmark datasets remain scarce for many fruit types.

Project Relevance: This survey supplies the field-level context needed to situate a papaya-specific project within the broader fruit-ripeness classification literature, and its explicit observation about dataset scarcity mirrors what several papaya-only studies above independently demonstrate. It can

be cited directly when framing the research gap and when justifying a deep-learning (CNN-based) methodological choice for the proposed project over purely statistical or spectroscopic alternatives.

2.7 PapayaFreshNet: A Hybrid Deep Learning Framework for Non-Destructive Freshness Classification of Papayas Using Convolutional and Transformer Networks

Sar, A., Choudhury, T., Sarkar, T., & Kotecha, K. (2025). PapayaFreshNet: A hybrid deep learning framework for non-destructive freshness classification of papayas using convolutional and transformer networks. Discover Food, 5(1), Article 97. https://doi.org/10.1007/s44187-025-00368-9

Summary: This 2025 study targets a related but distinct problem: classifying papaya freshness ("good" versus "bad/spoiled") rather than pre-harvest ripeness stage, motivated by the limits of plain CNNs in capturing both fine local texture and global spoilage patterns. The authors built a hybrid architecture combining a ResNet50 backbone for local feature extraction, a multi-head self-attention Transformer encoder for global dependencies, and Squeeze-and-Excitation blocks for channel-wise attention. Trained and tested on an augmented dataset of over 5,000 labelled papaya images and benchmarked against six other architectures (EfficientNetB2, XceptionNet, MobileNetV2, DenseNet121, a plain CNN, and InceptionV3), PapayaFreshNet achieved 97% accuracy, 93% precision, 94% recall, and a 0.97 ROC-AUC, with statistically significant improvements over the lighter baseline models confirmed through ANOVA and post-hoc testing.

Project Relevance: As the most recent (2025) papaya-related deep-learning study identified, this paper marks the field's current trajectory toward hybrid CNN-Transformer architectures and rigorous statistical validation, both of which are useful reference points for the proposed project's methodology and evaluation design. It also usefully delineates freshness detection (binary, post-harvest spoilage) from maturity classification (multi-class, pre-consumption ripeness stage), a distinction the proposed project should state explicitly when defining its own scope.

2.8 Computer Vision-Based Detection and Classification of Chemically Ripened Bananas and Papayas at Vendor Site Through Deep Learning AI Models Using Real-Time Dataset

Adhithya, S., & Kathirvelan, J. (2025). Computer vision-based detection and classification of chemically ripened bananas and papayas at vendor site through deep learning AI models using real-time dataset. Results in Engineering, 26, Article 104730. https://doi.org/10.1016/j.rineng.2025.104730

Summary: The problem addressed here is food safety: illegally, artificially accelerated ripening of fruits such as papaya (using agents like calcium carbide) poses health risks, and existing

laboratory-based detection methods are too slow for practical, on-site use by vendors or inspectors. The researchers built a custom real-time image dataset of naturally versus chemically ripened bananas (five species) and papaya, then benchmarked support vector machines, logistic regression, recurrent and artificial neural networks, standard CNNs, CNNs with Orthogonal Softmax Layers (OSL), and region-based CNNs. The best-performing model, CNN with OSL, achieved 91.58% accuracy for bananas and 96.32% accuracy for papaya, with inference in 2.24 seconds, and was successfully deployed on a low-cost Raspberry Pi with a camera for real-time, on-site operation.

Project Relevance: This paper demonstrates a fully deployable, low-cost, real-time computer-vision pipeline directly transferable to the kind of practical grading system the proposed project may aim to build, and its edge-device implementation offers a concrete deployment template. It also highlights chemically-induced ripening detection as an adjacent, food-safety-oriented application area worth acknowledging when framing why accurate, automated papaya classification matters beyond simple market grading.

2.9 Towards Precise Papaya Ripeness Assessment: A Deep Learning Framework with Dynamic Detection Heads

You, H., Fan, J., Huang, D., Yan, W., Zhang, X., Sun, Z., Liu, H., & Yuan, J. (2025). Towards precise papaya ripeness assessment: A deep learning framework with dynamic detection heads. Agriculture, 15(15), Article 1585. https://doi.org/10.3390/agriculture15151585

Summary: This study tackles the challenge of running accurate papaya ripeness assessment on resource-constrained edge devices for real-time, in-field orchard management, a constraint that most highly accurate models ignore. The authors propose ABD-YOLO-ting, a lightweight object-detection model built on YOLOv8n, combining a narrowed "YOLO-Ting" backbone, a low-computation ADown downsampling module, an enhanced BiFPN neck for efficient multi-scale feature fusion, and a dynamic detection head (DyHead) that improves sensitivity to the subtle colour and texture differences between maturity stages. On a self-constructed orchard dataset, the model achieved 94.7% mAP50 and 77.4% mAP50–95 with only 1.47 million parameters and 5.4 GFLOPs, outperforming mainstream YOLOv5, YOLOv8, and YOLOv11 baselines while remaining light enough for practical edge deployment.

Project Relevance: As the newest identified papaya-specific study, this paper marks a shift from lab-based, single-fruit classification toward real-time, in-field object detection, offering the proposed project both a lightweight-architecture reference and an important scope reminder: detecting and localizing fruit within a cluttered orchard scene is a distinct, complementary problem to classifying an already-isolated fruit image, and the proposed project should clearly state which of the two problems it addresses.

2.10 A Systematic Review and Comparative Meta-Analysis of Non-Destructive Fruit Maturity Detection Techniques

Rani, N., Garg, S., Bamel, K., Bhatt, V., Sharma, S., Mishra, S. K., Saini, N., & Parmar, S. (2024). A systematic review and comparative meta-analysis of non-destructive fruit maturity detection techniques. Plant Science Today, 11(1), 418–432. https://doi.org/10.14719/pst.2844

Summary: This review addresses the absence of a consolidated, quantitative comparison across the many disparate non-destructive fruit-maturity detection techniques scattered throughout the literature. The authors systematically reviewed 63 full-text articles published between 2012 and 2023, grouping techniques into three categories — machine vision (including deep learning), spectroscopy, and other miscellaneous methods — and conducted a formal meta-analysis using odds ratios, heterogeneity testing (I^2), and publication-bias checks (Egger's and Begg's tests). Machine vision, including deep-learning-based approaches, showed a positive correlation with reported accuracy and was identified as the most widespread, versatile, and low-cost option for high-speed maturity and visual-quality detection, while hyperspectral imaging was flagged as promising but costlier. The analysis also revealed high heterogeneity ($I^2 = 88.37\%$) across studies, indicating inconsistent methodologies and reporting standards field-wide.

Project Relevance: Although fruit-agnostic, this meta-analysis provides quantitative, cross-study evidence supporting a machine-vision/deep-learning approach as the most cost-effective and scalable choice for the proposed project, rather than costlier spectroscopic alternatives. Its explicit critique of methodological heterogeneity and inconsistent reporting also functions as a checklist of weaknesses — small datasets, non-standardized maturity definitions, limited statistical validation — that recur across the papaya-specific studies reviewed above and that the proposed project should deliberately avoid to produce a more rigorous and reproducible contribution.

3. Synthesis: Identified Gaps and Direction for the Proposed Project

Read together, these ten sources reveal three consistent patterns in the existing literature that define the opportunity for the proposed project.

1. Small, single-source, lab-controlled datasets. The majority of papaya-specific studies (Behera et al., 2020; Gayathri et al., 2021; Ratha et al., 2023) rely on the same order of magnitude of data — roughly 300 images captured at a single location under controlled lighting and a plain background. This limits confidence in how well reported accuracies (frequently 96–100%) would generalize to varied orchards, markets, lighting conditions, or papaya cultivars. A larger, more diverse, and more representative dataset is a clear and achievable contribution for the proposed project.

2. A trade-off between accuracy, cost, and deployability that papaya-specific research has only begun to resolve. The highest reported accuracies come either from costly sensing hardware,

such as hyperspectral imaging (Garillos-Manliguez & Chiang, 2021), or from computationally heavy hybrid architectures, such as the CNN-Transformer hybrid in PapayaFreshNet (Sar et al., 2025). Lightweight, field- or vendor-deployable approaches (You et al., 2025; Adhithya & Kathirvelan, 2025) are only just emerging for papaya-adjacent tasks such as orchard-level detection or chemical-ripening screening, leaving room for a project that explicitly optimizes for practical, low-cost deployment of maturity classification itself.

3. Inconsistent methodology, class definitions, and reporting standards across the field. The two broader survey and meta-analysis papers (Rizzo et al., 2023; Rani et al., 2024) independently confirm that fruit-maturity detection research as a whole suffers from scarce standardized datasets, high methodological heterogeneity, and uneven statistical validation — issues that are visible even within the papaya-specific literature reviewed here, where maturity is variously defined in two, three, five, or six classes with little cross-study consistency. The proposed project can strengthen its rigor and contribution by adopting a clearly justified, consistently applied maturity-class definition, reporting robust statistical validation (as modelled by Sar et al., 2025), and situating its dataset and results transparently against this existing body of work.

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Adhithya, S., & Kathirvelan, J. (2025). Computer vision-based detection and classification of chemically ripened bananas and papayas at vendor site through deep learning AI models using real-time dataset. Results in Engineering, 26, Article 104730. https://doi.org/10.1016/j.rineng.2025.104730

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